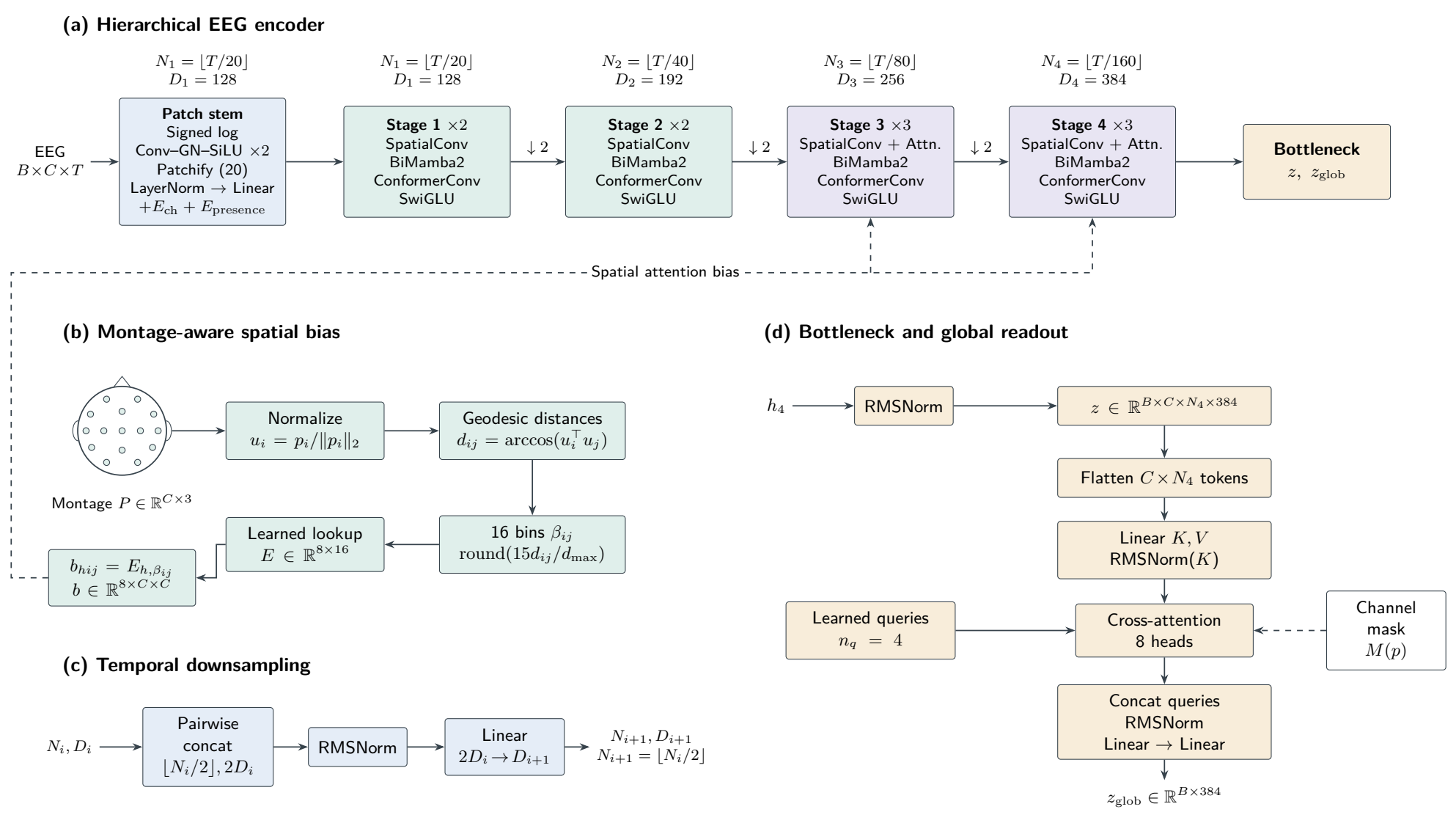

(a) Hierarchical EEG encoder
$N_1 = \lfloor T/20 \rfloor$
$D_1 = 128$
$N_1 = \lfloor T/20 \rfloor$
$D_1 = 128$
$N_2 = \lfloor T/40 \rfloor$
$D_2 = 192$
$N_3 = \lfloor T/80 \rfloor$
$D_3 = 256$
$N_4 = \lfloor T/160 \rfloor$
$D_4 = 384$
EEG
$B \times C \times T$
Patch stem
Signed log
Conv–GN–SiLU ×2
Patchify (20)
LayerNorm → Linear
$+E_{\text{ch}} + E_{\text{presence}}$
Stage 1 ×2
SpatialConv
BiMamba2
ConformerConv
SwiGLU
↓ 2
Stage 2 ×2
SpatialConv
BiMamba2
ConformerConv
SwiGLU
↓ 2
Stage 3 ×3
SpatialConv + Attn.
BiMamba2
ConformerConv
SwiGLU
↓ 2
Stage 4 ×3
SpatialConv + Attn.
BiMamba2
ConformerConv
SwiGLU
Bottleneck
$z,\ z_{\text{glob}}$
Spatial attention bias
(b) Montage-aware spatial bias
Montage $P \in \mathbb{R}^{C \times 3}$
Normalize
$u_i = p_i / \|p_i\|_2$
Geodesic distances
$d_{ij} = \arccos(u_i^\top u_j)$
16 bins β_{ij}
$\text{round}(15 d_{ij} / d_{\max})$
Learned lookup
$E \in \mathbb{R}^{8 \times 16}$
$b_{hij} = E_{h,\beta_{ij}}$
$b \in \mathbb{R}^{8 \times C \times C}$
(c) Temporal downsampling
N_i, D_i
Pairwise
concat
$\lfloor N_i/2 \rfloor, 2D_i$
RMSNorm
Linear
$2D_i \to D_{i+1}$
N_{i+1}, D_{i+1}
$N_{i+1} = \lfloor N_i/2 \rfloor$
(d) Bottleneck and global readout
h_4
RMSNorm
$z \in \mathbb{R}^{B \times C \times N_4 \times 384}$
Flatten $C \times N_4$ tokens
Linear K, V
RMSNorm(K)
Learned queries
$n_q = 4$
Cross-attention
8 heads
Channel
mask
$M(p)$
Concat queries
RMSNorm
Linear → Linear
$z_{\text{glob}} \in \mathbb{R}^{B \times 384}$